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Order Book Liquidity on Crypto Exchanges

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Abstract: We analyze intraday liquidity for a range of cryptocurrencies across different exchanges. Among the liquidity measures used, order book variation is most interesting for crypto traders, as it directly impacts their profit/loss. We find evidence that order book variation can be explained by liquidity measures indicating that trades are timed. We report various liquidity patterns that allow traders to increase their profits by minimizing liquidity-dependent trading costs. We further find indications that crypto exchanges can control liquidity by the number of offered currency pairs.

Keywords: cryptocurrencies; liquidity; market microstructure

1. Introduction

In recent years, cryptocurrencies have grown significantly in prominence. An increasing number of investors now recognize cryptocurrencies as a distinct asset class. This recognition has been fueled by the emergence of altcoins, many of which have outperformed Bitcoin in recent years, thereby expanding the investment opportunities within this new asset class. These developments have spurred considerable research on cryptocurrencies, as evidenced by a rapidly growing body of literature. However, much of this research remains centered on Bitcoin and focuses primarily on data derived from price series or trades data, such as returns and volatilities. Liquidity, despite being of critical importance to cryptocurrency investors, has received comparatively less attention. A key reason for this is the limited availability of data; while price data are readily accessible and trade data rather easy to obtain, order book data are more challenging to come by. Most current papers focus on liquidity measures on the trades level, neglecting to a large degree that cryptocurrency markets are mostly still not mature enough to provide the necessary deep and developed order books on lower levels than best-bid and best-ask that is needed for professional traders.

From an academic perspective, liquidity and its evolution over time serve as important indicators of, and prerequisites for, market efficiency. From an investor's perspective, liquidity directly affects transaction costs, which in turn influence the profitability of trading activities. An attractive trading platform provides the infrastructure necessary to trade (in our case, crypto) assets with high reliability and at low costs. This operational efficiency is a necessary, though not sufficient, condition for achieving higher forms of market efficiency.

A critical component in this context is the total transaction cost, which can be divided into explicit and implicit costs. Explicit costs include, but are not limited to, trading, clearing and settlement fees for exchanges, as well as service and commission fees for market intermediaries. These costs are typically straightforward to measure and compare due to their transparency. Implicit costs and risks, on the other hand, such as those arising

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from order book (il)liquidity, are not as transparent and are often significantly harder to quantify. While the intuitive notion of a negative relationship between liquidity and implicit trading costs holds, it remains somewhat vague as it does not pertain to a specific concept or aspect of liquidity. Different facets of liquidity give rise to various liquidity measures. Simple proxies, such as trading volume, may offer a rough indication of liquidity but fail to capture its more nuanced and specific aspects.

In this paper, we examine multiple dimensions of order book liquidity of cryptocurrencies. Specifically, we explore variations in liquidity across a large set of very diverse currency pairs, different exchanges and over time. The insights derived from this analysis hold value for academics, investors, trading venues and regulators.

For investors, comparing the liquidity and associated transaction costs of currency pairs and trading venues provides valuable insights to inform decisions about what to buy, where to buy it and when to execute trades. Trading platforms can leverage these results to reassess their trading systems, market design and listings. For instance, they might consider introducing designated sponsors for liquidity provision, transitioning from a purely order-driven market to a hybrid market. These liquidity providers play a crucial role in order book markets, particularly when liquidity for certain trading pairs is weak—e.g., when spreads are excessively large or volumes at the best levels of the order book are insufficient.

The concept of 24/7 continuous trading might also warrant reevaluation. For currency pairs with low liquidity, auction models or continuous trading interrupted by auctions could be more effective in bringing together a larger pool of traders. Increased liquidity attracts more participants, ultimately boosting trading volume and the market share of the exchange.

Finally, our findings are also pertinent for regulators. Market competition is a common concern for regulators. In a consolidated (fragmented) market, there is low (high) competition among exchanges but high (low) competition among traders, resulting in low (high) liquidity costs and high (low) explicit transaction costs. Achieving the optimal degree of market fragmentation is critical to safeguarding investor protection. In the crypto world, fragmentation occurs not only across exchanges but also within a single trading venue, as these typically offer numerous trading pairs with different base currencies.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 outlines the methodology and discusses the liquidity measures employed. Section 5 presents the results and Section 6 concludes.

2. Literature

2.1. Liquidity in Order Books

Research on market liquidity began in the late 1960s with Demsetz (1968), who examined bid-ask spreads in dealer markets. The subsequent literature has produced influential contributions across theoretical (e.g., Akerlof, 1970 for markets with information asymmetries; Kyle, 1985 for the strategic behavior of insiders), empirical (e.g., Huang & Stoll, 1996 on transaction costs in auction and dealer markets; Christie & Schultz, 1994 on collusion in the NASDAQ dealer market), and experimental domains (e.g., Plott & Sunder, 1982 on information dissemination in experimental auction markets; Friedman, 1993 on experimental comparisons of auction and dealer markets).

The present paper relates to several strands of the literature. Research on cryptocurrencies in finance and economics is still at an early stage. The majority of papers in this area focus on the potential real effects of cryptocurrencies as payment and transaction mechanisms. Ciaian et al. (2016), Harvey (2016), Böhme et al. (2015) and Raskin and Yermack (2018) provide broad perspectives on the economics of cryptocurrencies and the

underlying blockchain technology. Pagnotta and Buraschi (2018) propose valuation models for digital currencies. Easley et al. (2019) and Huberman et al. (2021) analyze bitcoin mining fees and the incentives of miners in equilibrium.

The literature examining cryptocurrency market liquidity has focused on centralized exchanges (CEX) or decentralized exchanges (DEX). While CEX employ limit order book markets, DEX usually use Automated Market Maker (AMM) protocols since limit order books are not feasable on-chain due to limited throughput and gas fees. Barbon and Ranaldo (2021) conducted a comparative study between the liquidity on centralized and decentralized exchanges and outline the rising competition of decentralized exchanges different concept of liquidity. Lehar and Parlour (2025) compare transaction-based liquidity measures between Uniswap v2 (DEX) and Binance (CEX). For CEXs, Brauneis et al. (2021) compare low-frequency and high-frequency liquidity measures for Bitcoin and Ethereum.

Brauneis et al. (2021) note that few studies utilize full order book data to examine the liquidity of cryptocurrency markets. Dyhrberg et al. (2018) investigate the trading dynamics of Bitcoin on US-based crypto exchanges, while Makarov and Schoar (2020) explore arbitrage opportunities across exchanges. In most cases, low-frequency measures have been employed without comparative frameworks. Brauneis and Mestel (2018) test the predictability of cryptocurrency returns, and Al-Yahyaee et al. (2020) study the efficiency of cryptocurrency markets, finding that inefficiencies are primarily driven by illiquidity and volatility in cryptocurrencies.

Anand et al. (2005), Back and Baruch (2007) and Ranaldo (2004) highlight the advantages of limit order books, which include, but are not limited to, low transaction costs (e.g., brokerage, commissions, spreads and fees), pre- and post-trade transparency, efficient price discovery through comprehensive information aggregation, reduced information asymmetry and a more competitive market microstructure.

The nature of liquidity provisioning in limit order book (LOB) markets differs significantly from that in quote-driven markets in one critical aspect. Limit order suppliers—i.e., the de facto market makers—face very low entry and exit barriers. They can freely provide or withdraw liquidity from the market based on prevailing market conditions. This contrasts with the designated market makers in quote-driven markets, who are obligated to provide liquidity. This key feature of liquidity provisioning in LOB markets is encapsulated by the "free-entry and free-exit" hypothesis proposed by Brockman and Chung (2002).

Previous literature has extensively documented the informativeness of limit order books and their role in reducing information asymmetry. Cao et al. (2009) examined the informational content of order book levels beyond the best bid and ask. Similarly, L. E. Harris and Panchapagesan (2005) find that limit order books provide valuable insights into future price movements. The state of the order book—characterized by tightness, depth and imbalance—can be assessed using various measures, such as spread, volume and impact costs.

2.2. Liquidity Measures in Order Books

Liquidity in limit order book (LOB) markets is a dynamic interplay between market orders, which consume liquidity, and limit orders, which supply it. Traders actively monitor the market and engage in competitive behaviors to optimize their trade execution by either demanding or supplying liquidity (S.-L. Chung et al., 2012; Foucault, 1999; L. Harris, 1998; Parlour, 1998). This endogeneity in traders' decisions reflects their attempts to adapt to prevailing market conditions. For instance, traders frequently undercut existing quotes to improve upon the best bid-ask prices, showcasing the competitive dynamics of liquidity supply and demand (Biais et al., 1998).

Impatient traders typically employ market orders to consume liquidity, while patient traders place limit orders, thereby supplying liquidity. These limit order traders are compensated for their patience and for providing the service of liquidity; however, they also bear execution risk and pick-off risk from informed traders (Baruch, 2005; Seppi, 1997). The LOB structure efficiently matches liquidity suppliers with demanders, maintaining a balance that is critical for market operations.

The relationship between liquidity supply and demand is influenced by market conditions. Research has shown that limit orders increase significantly during periods of low liquidity, as indicated by widened spreads and other liquidity metrics (K. Chung et al., 1999; Griffiths et al., 2000; L. Harris & Hasbrouck, 1996; Hollifield et al., 2004; Ranaldo, 2004). Conversely, during periods of high liquidity, the proportion of market orders rises. This dynamic creates a mean-reverting behavior in bid-ask spreads and contributes to the ecological equilibrium in liquidity supply and demand within the market (Handa et al., 2003, 1998; Handa & Schwartz, 1996).

The nature of liquidity provisioning in LOB markets contrasts sharply with that in quote-driven markets. In LOBs, liquidity suppliers face minimal entry and exit barriers, allowing them to adapt quickly to market conditions. This "free-entry and free-exit" characteristic, as described by Brockman and Chung (2002), is a defining feature of LOB markets. However, its implications for market resilience and reliability, particularly during crises, remain underexplored (Anand & Venkataraman, 2016; Tripathi et al., 2021).

Recent studies have highlighted asymmetric behavior between the bid and ask sides of LOBs, particularly during crisis periods. For instance, (Cenesizoglu & Grass, 2018) found that the ask side deteriorates more significantly than the bid side during crises, contributing disproportionately to market instability. (Sensoy, 2019) also noted these asymmetries, emphasizing the need for separate analyses of the bid and ask dimensions of liquidity.

Most liquidity studies in LOBs focus on spread and depth, derived from price and volume information at the best bid-ask levels (Chordia et al., 2000, 2002, 2008). However, evidence from developed markets suggests that the best bid-ask orders may be noisy and less informative (Amaya et al., 2018; Cao et al., 2009; Cenesizoglu & Grass, 2018; Deuskar & Johnson, 2011). Deeper levels of the order book often provide a more accurate assessment of liquidity, but accessing such data is challenging, particularly in emerging markets (Bekaert et al., 2007; Lesmond, 2005).

The components of the bid-ask spread also differ significantly between dealer and LOB markets. In dealer markets, spreads primarily compensate designated market makers for inventory holding, information asymmetry and order processing costs (Amihud & Mendelson, 1986; Glosten & Milgrom, 1985, Ho & Stoll, 1983). In contrast, in LOBs, spreads reflect transaction costs, adverse selection costs and waiting costs borne by limit order suppliers (Brockman & Chung, 1999; Chan, 2000; de Jong et al., 1996).

The shape of the LOB, characterized by parameters such as depth, height, slope, resilience and elasticity, significantly impacts price formation, liquidity provisioning and volatility (Cao et al., 2009; Næs & Skjeltorp, 2006). A thick order book with depth concentrated around the best quotes reflects a consensus on the true price and promotes stability (Goettler et al., 2005; Valenzuela et al., 2015). Conversely, thin order books with dispersed depth indicate mispricing and increased short-term volatility (Obizhaeva & Wang, 2013; Roşu, 2009, 2020). Asymmetries in the order book shape can signal higher trading costs and potential inefficiencies (Biais et al., 1995).

The resilience of the order book—its ability to recover from order flow shocks—is a critical yet underexplored aspect of market efficiency (Degryse et al., 2005; Large, 2007). Recent advancements in high-frequency data availability provide opportunities to examine this property more closely, particularly during crisis events.

Future research could explore the resilience of LOBs in greater detail, particularly examining how free-entry and free-exit dynamics affect liquidity provisioning during market stress. Additionally, asymmetries between the bid and ask sides warrant further investigation to understand their implications for market stability. Finally, expanding the study of liquidity measures to emerging markets, where data limitations persist, could provide valuable insights into the global dynamics of liquidity.

This paper contributes to the literature in several ways. First, we focus on liquidity, whereas most prior studies on cryptocurrencies have analyzed other aspects. Second, instead of limiting our analysis to Bitcoin, we examine a broad set of altcoins. Additionally, we contribute to the sparse body of research that incorporates deeper levels of order books—considering prices and quantities beyond the best bid and ask—when assessing limit order book liquidity. Third, unlike order books from other markets, the full limit order books of cryptocurrency exchanges are unique in providing 24/7 trading data.

3. Data

3.1. Raw Data

Order book data from the exchanges Binance, Kraken, Huobi and OKEx were obtained through CryptoTick (https://www.cryptotick.com/, accessed on 6 August 2021). The sample period spans 1 January 2019, to 30 September 2019, covering 273 days. The data are provided as individual files for each currency pair, exchange and trading day, with dates and times standardized to Coordinated Universal Time (UTC).

The dataset includes all levels from both sides of the order book as made available to CryptoTick by the exchanges. While some exchanges report the full set of orders, others impose limitations. For instance, Binance reported only the top 20 bids and asks until mid-June 2019.

The time frame of the sample captures numerous new listings and delistings on the exchanges, with some currency pairs appearing or disappearing between January and September 2019. Additionally, certain trading days are missing from the dataset, likely due to suspended trading or unavailability of data for unknown reasons.

These 24/7 limit order book data enable the reconstruction of (parts of) the order books for evaluating the liquidity of currency pairs. Unlike traditional markets, where trading hours are limited, the continuous trading nature of cryptocurrency markets may shift or concentrate liquidity at specific times of the day. Notably, trading activity captured in our limit order books is uninterrupted by auctions, in contrast to systems like XETRA, which employs opening, intraday and closing auctions for German stocks.

3.2. Data Processing

Each file begins with an order book snapshot that lists all available price levels on both sides of the order book at the start of the day. Subsequent entries document updates to the current order book status. To facilitate analysis, we created 5-min order book snapshots containing up to 20 levels on each side (if available), resulting in 288 order book snapshots per day. The use of 5-min intervals is widely accepted in the literature as it represents a balanced compromise between computational efficiency and the retention of essential informational content.

To ensure data quality, we applied several filters. First, we excluded all days with order books reporting negative or zero bid-ask spreads at five-minute intervals. Such instances should have triggered the matching and removal of the corresponding orders from the order book. Second, we omitted all days with order books reporting negative quantities for any order, as this is economically nonsensical.

These issues indicate inaccuracies in the raw data. Since we could not identify where the errors in the raw data occurs, we omitted the entire day's observations for that pair. Such omissions are rare in our sample and are unlikely to significantly impact our results. This filter applies to 2543 dates across 453 trading pairs (1392 on OKeX, 767 on Binance, 324 on Huobi and 60 on Kraken), less than 2% of the raw data, resulting in 254,728 observations across 1086 trading pairs.

3.3. Descriptive Statistics

The crypto exchanges in our sample offer a variety of cryptocurrency trading pairs. For this paper, we define the cryptocurrency reported as the quantity in the order book as the "target currency," and the cryptocurrency or fiat currency reported as the price in the order book as the "base currency." Each combination of a target currency with a base currency constitutes a currency pair (or trading pair).

Table 1 presents, for each exchange, the number of base currencies, target currencies and trading pairs traded between January 2019 and September 2019. While the number of base currencies across exchanges is relatively modest, ranging between four and twelve, there are significant differences in the number of target currencies and trading pairs. Binance and OKEx offer a broader selection, with 162 and 160 target currencies, resulting in 515 and 436 trading pairs, respectively. In contrast, Huobi and Kraken adopt a more focused approach, offering 22 and 21 target currencies (57 and 79 trading pairs, respectively).

Table 1. Number of base currencies, target currencies and trading pairs by exchange. This table reports the number of base currencies (column 2), the number of target currencies (column 3) and the number of trading pairs (column 4) by the four exchanges (column 1).

Exchange	# of Base Currencies	# of Target Currencies	# of Trading Pairs
BINANCE	11	161	514
HUOBI	4	22	57
KRAKEN	7	21	79
OKEX	5	160	436

These differences highlight varied business policies or strategies among crypto exchanges: Binance and OKEx prioritize a wider selection of trading pairs, while Huobi and Kraken focus on specialization within a more limited set of currency pairs.

Next, we categorize the base currencies into three types: fiat currencies (e.g., US dollar, Euro), stablecoins (e.g., Tether and other US dollar-pegged coins) and other cryptocurrencies (e.g., Bitcoin, Ether).

Table 2 presents the number of target currencies and trading pairs by exchange and by our selected base currency types. Among the exchanges, Huobi is the only one to offer all three base currency types. In contrast, Binance and OKEx do not support fiat currencies, while Kraken does not offer any stablecoins.

Huobi's four fiat trading pairs involve the US dollar paired with Tether, Bitcoin, Ether and Ripple. These four currencies are widely used and serve as base currencies, making these pairs convenient entry and exit points between the fiat and cryptocurrency markets. The other three exchanges focus on providing either stablecoins or fiat currencies as base currency options.

Table 2. Number of target currencies and trading pairs by exchange and base type. This table reports the number of target currencies (column 3) and the number of trading pairs (column 4) by the four exchanges (column 1) and the base currency type (column 2).

Exchange	Base Type	# of Target Currencies	# of Trading Pairs
BINANCE	cryptocurrency	157	377
BINANCE	stablecoin	57	137
HUOBI	cryptocurrency	20	33
HUOBI	fiat currency	4	4
HUOBI	stablecoin	20	20
KRAKEN	cryptocurrency	20	29
KRAKEN	fiat currency	19	50
OKEX	cryptocurrency	140	276
OKEX	stablecoin	143	160

Since the exchanges differ in the types of base currencies offered, we now examine which base currencies are available on each exchange and highlight the most notable findings. Table 3 summarizes this information, with the first column listing the exchanges, the second column showing the base currencies associated with each exchange, the third column indicating the type of the base currency and the fourth column reporting the number of target currencies for each base currency. The total number of currencies traded during our sample period (including new listings and delistings) is also presented.

Table 3. Number of target currencies by exchange and base currency. This table reports the number of target currencies (column 4) by the four exchanges (column 1) and the base currencies (column 2). Column 3 defines the type of the corresponding base currency in column 2.

Exchange	Base Currency	Base Type	# of Target Currencies
BINANCE	BBTC	cryptocurrency	1
BINANCE	BNB	cryptocurrency	92
BINANCE	BTC	cryptocurrency	150
BINANCE	ETH	cryptocurrency	131
BINANCE	TRX	cryptocurrency	1
BINANCE	XRP	cryptocurrency	2
BINANCE	PAX	stablecoin	29
BINANCE	TUSD	stablecoin	24
BINANCE	USDC	stablecoin	27
BINANCE	USDS	stablecoin	2
BINANCE	USDT	stablecoin	55
HUOBI	BTC	cryptocurrency	20
HUOBI	ETH	cryptocurrency	13
HUOBI	USD	fiat currency	4
HUOBI	USDT	stablecoin	20
KRAKEN	BTC	cryptocurrency	20
KRAKEN	ETH	cryptocurrency	9
KRAKEN	CAD	fiat currency	7
KRAKEN	EUR	fiat currency	18
KRAKEN	GBP	fiat currency	2
KRAKEN	JPY	fiat currency	4
KRAKEN	USD	fiat currency	19
OKEX	BTC	cryptocurrency	127
OKEX	ETH	cryptocurrency	118
OKEX	OKB	cryptocurrency	31
OKEX	USDK	stablecoin	18
OKEX	USDT	stablecoin	142

Bitcoin and Ether emerge as the cryptocurrencies associated with the largest number of trading pairs, particularly on Binance and OKEx, where they are part of over 100 trading pairs. Tether, as a base currency, is used for 142 target currencies on OKEx, compared to 55 on Binance and 20 on Huobi. Binance additionally offers other stablecoins and cryptocurrencies, such as Baby Bitcoin (BBTC), Tron (TRX), Ripple (XRP) and USD Coin (USDS), though these are of limited significance and will not be considered further in this paper. Both Binance and OKEx have introduced their own coins—Binance Coin (BNB) with 92 target currencies and OKB with 31 target currencies—adding to the extensive range of investment options. This variety, while beneficial, can be challenging for investors to fully comprehend and evaluate.

Kraken stands out as the only exchange in our sample that offers fiat currencies in addition to the US dollar (or USD-pegged stablecoins). The fiat currencies provided by Kraken include the euro (EUR), Canadian dollar (CAD), Japanese yen (JPY) and Pound Sterling (GBP). These offerings may appeal to investors from corresponding regions, as they allow target currencies to be valued directly in their home fiat currency, thereby eliminating currency risk and transaction costs associated with converting to and from the US dollar. Overall, US dollar-based investors benefit from the wider range of opportunities offered by fiat trading pairs and USD-pegged stablecoins.

Table 4 shows the distribution of trading pairs and target currencies across the four exchanges in our sample. Approximately 72% of the target currencies and 82% of the trading pairs are traded on only one exchange. Conversely, only eleven target currencies (4.26%) and twelve trading pairs (1.37%) are available on all four exchanges. This limited overlap may be inconvenient for investors, as it requires them to maintain multiple wallets to trade across a variety of crypto assets.

Table 4. Number of trading pairs and target currencies traded parallel. This table summarizes the number of target currencies (panel (a)) and trading pairs (panel (b)) that are traded simultaneously across the four exchanges in our sample. The first row in each panel provides the absolute numbers, while the second row shows the corresponding relative frequencies (percentages). Columns 1 through 4 indicate the number of exchanges on which the target currencies or trading pairs are available, ranging from one exchange (column 1) to all four exchanges (column 4).

(**a**) **Target Currencies**					(**b**) **Trading Pairs**				
	1	**2**	**3**	**4**		**1**	**2**	**3**	**4**
Frequency	182	54	10	11	Frequency	714	117	30	12
Rel. Freq.	70.8%	21.0%	3.9%	4.3%	Rel. Freq.	81.8%	13.4%	3.4%	1.4%

Restricting the comparative evaluation of exchanges to trading pairs shared by all exchanges could introduce bias into the results. For instance, Binance and OKEx, with their broader range of trading pairs, likely have their total liquidity distributed across a greater number of pairs. This could make an individual trading pair from Binance or OKEx appear less liquid compared to the corresponding pair on Huobi or Kraken, which offer fewer trading pairs.

From an investor's perspective, this issue is only significant for those who exclusively trade these overlapping pairs. To better understand the overall market and appeal to a wider audience, our approach evaluates liquidity by analyzing all trading pairs available on the exchanges and applying appropriate liquidity measures, irrespective of the target currency, base currency or specific trading pairs.

4. Methodology

A significant challenge in analyzing data from cryptocurrency exchanges lies in the sheer number and diversity of trading pairs, as discussed in Section 3.3. These trading pairs vary widely across several dimensions, such as price level, transaction volume, exchange rate risk and other characteristics. This heterogeneity complicates the selection of liquidity measures that can be meaningfully compared across such a broad set of trading pairs.

To address this challenge, we select liquidity measures that can be applied universally across trading pairs and ensure their comparability, regardless of the target or base currency. Measures that inherently depend on the units of either the target or base currencies—such as those based on trading volume—are excluded. Trading volume, for example, can be expressed either as the quantity of target currency units traded or as the transaction size, calculated by multiplying the trade volume by the average trade price. These approaches are problematic for our analysis because they are not easily comparable over time or between heterogeneous assets.

Additionally, measures for ex-ante trading costs are not suitable for our methodology. Approaches like those proposed by Gomber et al. (2015) and Brauneis et al. (2021) require all trading pairs to share the same base currency. Given that our sample includes a variety of target and base currencies across multiple exchanges, implementing such methods is infeasible. Furthermore, the varying levels of interest in different target currencies lead to significant disparities in trading volumes. As a result, liquidity measures based on trading volumes of a single base currency across exchanges would disproportionately favor popular currencies, making unpopular coins appear less liquid in most cases. Limiting comparisons to identical trading pairs across exchanges would yield minimal insights into the overall liquidity of cryptocurrency exchanges.

Since cryptocurrency markets operate 24/7 and our data are sampled at five-minute intervals, the resulting liquidity measures at this frequency would be prone to high levels of noise. To mitigate this, we compute daily averages of the liquidity measures, which provide more stable and reliable insights compared to measures derived from single, high-frequency order book snapshots.

Our analysis begins by comparing liquidity across trading pairs based on their base currency types—fiat currencies, stablecoins or other cryptocurrencies. We group the results for our selected liquidity measures by base currency types, as presented in Table 2, and also examine specific base currencies, as shown in Table 3. We exclude less relevant currencies, such as BBTC, TRX, XRP and USDS on Binance, as they account for only a small fraction of the target currencies for their respective base currency types.

The following subsection presents and discusses the liquidity measures used in this study. All selected measures are derived solely from order book data, particularly from prices and quantities across various order book levels, which reflect liquidity provision by market participants.

4.1. Order Book Spread

Order book spread measures the liquidity implied by the prices of orders at the best bid and ask levels. The lowest-priced sell limit order represents the best ask price, while the highest-priced buy limit order represents the best bid price. The difference between these two prices is known as the quoted spread. K. H. Chung et al. (2004) highlight that investors submit limit orders to the order book based on the expected profitability of their orders. Given the high volatility inherent in cryptocurrencies, wider spreads can be anticipated to account for this price volatility.

In general, the bid-ask spread reflects both explicit costs, such as transaction and order processing costs, and implicit costs, including adverse selection and waiting costs. Brockman and Chung (1999), Chan (2000) and de Jong et al. (1996) find that a significant portion of the spread is persistent and attributable to adverse selection costs. Furthermore, Brauneis et al. (2021) emphasize that order book data at the best bid and ask levels provide reliable, real-time estimates of liquidity in cryptocurrency markets.

Quoted spreads are expressed in terms of prices and, in our dataset, are denominated in the base currency. To ensure comparability across currencies and exchanges, we compute the relative spread by dividing the quoted spread by the mid-quote. This normalization removes the base currency denomination, making the results uniform and comparable across the diverse trading pairs in our sample.

$$\text{Relative Spread} = \frac{P_{\text{best ask}} - P_{\text{best bid}}}{P_{\text{mid}}} = \frac{\text{Quoted Spread}}{P_{\text{mid}}} \tag{1}$$

with

$$P_{\text{mid}} = \frac{P_{\text{best ask}} + P_{\text{best bid}}}{2} \tag{2}$$

Gomber et al. (2015) interpret the relative spread as a liquidity premium that investors must pay to execute an order immediately. Investors face a trade-off between paying this liquidity premium and incurring the waiting costs associated with delayed execution.

Up to this point, we have focused exclusively on the best bid and ask prices. However, the literature, including Cao et al. (2009), provides evidence that deeper levels in the order book are less affected by noise and often contain more meaningful information about liquidity in limit order books. To incorporate the impact of prices and volumes beyond the best bid and ask levels, volume-weighted average prices (VWAP) can be employed. The VWAP for the bid and ask sides are calculated as follows:

$$P_L^{\text{VWAP}} = \frac{\sum_{l=1}^{L} P_l * Q_l}{\sum_{l=1}^{L} Q_l}, \tag{3}$$

where l represents the levels of the corresponding bid and ask sides, Q_l equals the quantity at level l and P_l equals the price at level l.

The VWAP spread and the VWAP relative spread at level L are calculated analogously to the (best) bid-ask spread:

$$\text{Spread}_L^{\text{VWAP}} = P_{\text{ask},L}^{\text{VWAP}} - P_{\text{bid},L}^{\text{VWAP}} \tag{4}$$

$$\text{Relative Spread}_L^{\text{VWAP}} = \frac{P_{\text{ask},L}^{\text{VWAP}} - P_{\text{bid},L}^{\text{VWAP}}}{P_{\text{mid}}} \tag{5}$$

Besides level one, the best bid and ask level, we calculate the VWAP spreads for the levels 5, 10, 15 and 20, i.e., incorporating up to the best 20 orders on both sides of the order book if available.

4.2. Order Book Depth

The liquidity impact of orders at the best prices is greater and more persistent when these orders carry substantial volume. Cao et al. (2009) point out that orders at the best prices are often noisy and less informationally efficient than prices at deeper levels of the order book. This inefficiency arises because these orders are matched first and disappear when market orders with larger volumes surpass the best orders, or when new limit orders with smaller volumes undercut the existing best orders. As a result, the best prices and quantities tend to be more volatile and less stable over time.

To mitigate the impact of this noise, we rely on five-minute order book snapshots and calculate daily averages. This approach allows us to smooth out transient fluctuations and obtain an unbiased representation of the liquidity state of the order book.

Goettler et al. (2005) and Valenzuela et al. (2015) argue that greater depth around the best price levels reflects a stronger consensus on the true price, while a concentration of volume at price levels further away from the mid-price indicates greater uncertainty about the price.

We define depth as the sum of the product of the price and quantity from the best price level to a specified depth level. To analyze differences between buy and sell liquidity, we calculate depth separately for both sides of the order book as follows:

$$\text{Depth}_{\text{ask},L} = \sum_{l=1}^{L} (P_{\text{ask},l} * Q_{\text{ask},l}) \tag{6}$$

$$\text{Depth}_{\text{bid},L} = \sum_{l=1}^{L} (P_{\text{bid},l} * Q_{\text{bid},l}) \tag{7}$$

This depth measure captures the accumulated volume present in the order book across different price levels. By taking the product of quantity and price, we normalize the results to the base currency, effectively canceling out the target currency. This approach enables a consistent comparison of depth across exchanges for a given base currency.

We avoid comparing results in terms of the target currency, as the sheer number of target currencies and their limited overlap across exchanges—most being traded exclusively on a single exchange—would make such comparisons impractical and less meaningful.

4.3. Order Book Variation

Next, we propose a new measure to capture order book variation. The underlying idea is that one characteristic of liquid markets is their stability over time: order books in such markets typically do not change rapidly because they can absorb trading demand without significant disruptions. While periods of rapid changes in prices and order books can occur, we argue that, on average, the order books of liquid markets should exhibit less variation than those of illiquid markets.

As outlined in Section 3, we reconstruct the order book at five-minute intervals, resulting in 288 snapshots per day. To derive a comparable and non-price-dependent measure, we count the number of order book levels by which the mid-price moves from one snapshot to the next. Specifically, we identify the level l that minimizes the absolute difference between the bid price at level l at time t and the mid-price at time $t-1$.

$$l_{\text{bid}} = \arg\min_{l}(|P_{bid,1,t} - P_{mid,t-1}|, |P_{bid,2,t} - P_{mid,t-1}|, ..., |P_{bid,l,t} - P_{mid,t-1}|, ..., |P_{bid,L,t} - P_{mid,t-1}|) \tag{8}$$

Next, we proceed analogously for the ask side of the order book:

$$l_{\text{ask}} = \arg\min_{l}(|P_{ask,1,t} - P_{mid,t-1}|, |P_{ask,2,t} - P_{mid,t-1}|, ..., |P_{ask,l,t} - P_{mid,t-1}|, ..., |P_{ask,L,t} - P_{mid,t-1}|) \tag{9}$$

We combine the results from Equations (8) and (9) by taking the maximum of their results:

$$l = \max(l_{\text{bid}}, l_{\text{ask}}) \tag{10}$$

By construction, at least one of the two values must be greater than or equal to one. Lower values of l indicate lower variation and hence higher relative liquidity in the order book.

For each trading day, we calculate 288 values of l (one value for each five-minute interval). To facilitate further analysis, we aggregate these values in two ways. First, we compute the simple average of l across all observations for each trading day. This average filters out short-term fluctuations and provides a measure of the average level of order book variation for a currency pair on a given trading day. Second, we calculate the maximum five-minute variation, which represents the worst-case variation recorded for a currency pair on a given trading day. The difference between these two metrics serves as a straightforward measure of variation risk.

This variation measure captures two key factors that consume liquidity simultaneously. On one hand, best bid and ask orders may be canceled and removed from the order book. From a narrow order book liquidity perspective, this constitutes a removal of liquidity. However, we expect the impact of cancellations on mid-price changes to be minimal, as there is no evidence in the literature, to the best of our knowledge, suggesting that order cancellations significantly drive mid-price variation.

On the other hand, liquidity can be consumed through trades, as matched orders disappear from the order book. The impact of a trade depends on its size and the liquidity available in the order book at the time. Larger trades and smaller order sizes at levels close to the mid-price are likely to result in higher values for our variation measure. Trades are thus expected to be the primary driver of significant order book variation.

One advantage of this measure is its ability to capture the dynamics of order book liquidity without overly emphasizing exact price values. By abstracting from prices and reducing the information to price levels, this measure facilitates comparisons across trading pairs with different base currencies. From an investor's perspective, variation is highly relevant, as it can help identify trading pairs with low liquidity. Investors can use this information to adjust their order submission strategies, minimizing variation, execution and pick-off risk.

Closely related to our measure of order book variation is the volatility since level movements in the order book leads to returns. This allows us to compare if order book variations and return volatility actually measure the same aspect of liquidity or yield to different results. Moreover, our data allows to calculate return volatility from a five-minute interval compared to volatility based on daily returns from public sources (e.g., coinmarketcap.com) We measure volatility as the five-minute annualized log return volatility.

$$\sigma = \sqrt{Variance(ln(\frac{P_t}{P_{t-1}})) * 288 * 365} \tag{11}$$

with P_t is the price at time t and P_{t-1} is the price 5 min before time t. To annualize the volatility, we need to use the factor 288 since we have 288 five-minute intervals during a day and the factor 365 since trading in crypto is possible during 365 days in a year. For each trading pair and date, we receive the annualized return volatility based on the intraday five-minute returns.

4.4. Order Book Imbalance

While depth itself captures an important aspect of liquidity, the difference between bid and ask depth—referred to as *order imbalance*—provides additional valuable information. Biais et al. (1995) present evidence that a higher order imbalance is associated with increased trading costs. Similarly, Bonart and Gould (2017) argue that order book imbalance is a strong predictor of order flow, as traders may cancel their limit orders and replace them with market orders in response to imbalances.

We calculate order imbalance by subtracting the bid depth from the ask depth at the selected levels of the order book.

$$\text{OBI}_L = \sum_{l=1}^{L} (P_{\text{ask},l} * Q_{\text{ask},l} - P_{\text{bid},l} * Q_{\text{bid},l}) = \text{Depth}_{\text{ask},L} - \text{Depth}_{\text{bid},L} \tag{12}$$

Similar to the spread, normalization is necessary to ensure that the order imbalance measure is comparable across different base currencies. To achieve this, we divide the order imbalance—calculated as the difference between ask and bid depths—by the sum of the ask and bid depth measures at the same order book levels.

$$\text{NOBI}_L = \frac{\sum_{l=1}^{L} (P_{\text{ask},l} * Q_{\text{ask},l} - P_{\text{bid},l} * Q_{\text{bid},l})}{\sum_{l=1}^{L} (P_{\text{ask},l} * Q_{\text{ask},l} + P_{\text{bid},l} * Q_{\text{bid},l})} = \frac{\text{OBI}_L}{\text{Depth}_{\text{ask},L} + \text{Depth}_{\text{bid},L}} \tag{13}$$

This measure addresses several limitations of order book depth. First, it eliminates the influence of target and base currencies, enabling comparisons across all base currencies and exchanges. Second, the order book imbalance measure is standardized to values between -1 and 1, making it straightforward to interpret. A value of -1 represents the extreme case where the order book contains only bid orders, while 1 represents the opposite extreme with only ask orders. A value of 0 indicates a balanced order book, with equal depth at level L for both sides.

Accordingly, a negative normalized order book imbalance (NOBI) reflects a dominance of bid orders in terms of the base currency's value, whereas a positive NOBI indicates a predominance of ask orders.

To measure the effect of order book imbalance in general, we take the absolute value of NOBI for the absolute normalized order book imbalance (ANOBI):

$$\text{ANOBI}_L = |NOBI_L| \tag{14}$$

5. Results

In this section, we present the results of our analysis, focusing on the liquidity characteristics of cryptocurrency markets across different exchanges and base currencies. The results are organized into subsections that examine key liquidity measures, including spreads, order book depth, imbalance and variation. Additionally, we explore the interplay between these measures and how they vary across trading pairs, exchanges and time. Our findings provide insights into the unique features of cryptocurrency markets, the factors driving liquidity differences and the implications for market participants. The subsequent subsections detail these results, highlighting both aggregate trends and exchange-specific observations.

5.1. Spreads

Table 5 presents the average relative spreads at various order book levels—best level, level 5, level 10, level 15 and level 20—grouped by exchange and base currency type. Several notable differences emerge between the exchanges and base currency types.

For cryptocurrencies other than stablecoins, the average spreads are approximately 0.55% on Binance, Huobi and Kraken, but significantly higher at 1.9% on OKEx. Substantial differences are also observed for stablecoins, with average spreads of 0.59% on Binance, 0.23% on Huobi and 1.26% on OKEx. For fiat currencies, the spreads vary considerably between Huobi (0.12%) and Kraken (1.25%), with the differences primarily driven by higher spreads for GBP, JPY and CAD on Kraken. Even for USD, the average spread at the

best level on Kraken (0.38%) remains significantly higher than the corresponding spread on Huobi.

Table 5. Spread measures by exchange and base currency. This table reports the average relative spreads in percent (%) for the best level (column 3), level 5 (column 4), level 10 (column 5), level 15 (column 6) and level 20 (column 7) by the four exchanges (column 1) and base currency types (column 2). The numbers in the brackets report the standard errors of the corresponding average relative spreads.

Exchange	Base Type	L1 Rel. Spread	L5 Rel. Spread	L10 Rel. Spread	L15 Rel. Spread	L20 Rel. Spread
BINANCE	cryptocurrency	0.552 (0.003)	1.53 (0.012)	2.78 (0.023)	4.21 (0.037)	5.92 (0.0558)
BINANCE	stablecoin	0.589 (0.018)	1.14 (0.031)	2.62 (0.0832)	4.9 (0.153)	7.37 (0.226)
HUOBI	cryptocurrency	0.539 (0.008)	1.17 (0.019)	1.99 (0.0358)	3.38 (0.0886)	4.76 (0.126)
HUOBI	fiat currency	0.12 (0.004)	0.283 (0.010)	0.622 (0.0268)	1.04 (0.0426)	1.44 (0.0575)
HUOBI	stablecoin	0.231 (0.005)	0.459 (0.010)	0.769 (0.0163)	1.12 (0.0254)	1.52 (0.0367)
KRAKEN	cryptocurrency	0.576 (0.0112)	1.19 (0.024)	2.18 (0.0426)	3.3 (0.0644)	4.44 (0.0875)
KRAKEN	fiat currency	1.25 (0.044)	2.05 (0.057)	4 (0.122)	7.15 (0.202)	12.4 (0.434)
OKEX	cryptocurrency	1.9 (0.052)	4.76 (0.094)	11.3 (0.513)	17.1 (0.698)	21.2 (0.75)
OKEX	stablecoin	1.26 (0.053)	3.12 (0.105)	6.73 (0.585)	9.29 (0.295)	11.9 (0.383)

The results indicate that specialization in selected trading pairs enhances overall exchange liquidity. Binance and OKEx offer 514 and 436 trading pairs, respectively, while Huobi and Kraken provide only 57 and 79 trading pairs. Despite Kraken's larger best relative spread for cryptocurrencies (other than stablecoins) compared to Binance, Kraken and Huobi show higher liquidity for these cryptocurrencies at deeper levels, such as level 5, as indicated by lower relative spreads compared to Binance and OKEx.

Given that Binance supports 377 trading pairs for cryptocurrencies, compared to only 33 for Huobi and 29 for Kraken, the liquidity of Binance in terms of spreads is surprisingly competitive for other cryptocurrencies. This is particularly evident when considering the higher concentration of liquidity on Huobi.

The contrast becomes even more apparent when comparing the relative spreads of stablecoins across exchanges. Huobi's stablecoins exhibit significantly lower relative spreads compared to those on Binance and OKEx, highlighting a more concentrated liquidity. Although Binance and OKEx adopt similar strategies by offering a large number of trading pairs in cryptocurrencies and stablecoins, their liquidity, as measured by relative spreads, varies substantially.

The spreads increase with higher levels of the order book, and the steeper the increase in spreads, the higher the spreads at the best level. Among the exchanges and base currency types, Huobi stands out with a notably flat increase in spreads across order book levels for fiat currencies and stablecoins.

A strong relationship between relative spreads across order book levels is observed, which is partly expected due to the nested structure of the spread measures (higher levels

inherently include lower levels). Table 6 presents the correlations among spreads at different order book levels, providing further support for these findings.

Table 6. Correlation between spread measures. This table reports the correlation matrix between level 1 relative spread, level 5 relative spread, level 10 relative spread, level 15 relative spread and level 20 relative spread. All correlations are significantly different from zero at the 1%-significance level.

	L1 Rel. Spread	L5 Rel. Spread	L10 Rel. Spread	L15 Rel. Spread
L1 Rel. Spread				
L5 Rel. Spread	0.81 ***			
L10 Rel. Spread	0.57 ***	0.61 ***		
L15 Rel. Spread	0.50 ***	0.64 ***	0.86 ***	
L20 Rel. Spread	0.56 ***	0.67 ***	0.77 ***	0.86 ***

*** indicates 0.01 significance level for a two-tailed test.

Table 7 presents the correlations between spread measures and order book variation measures. All correlations are negative and statistically significant: higher spreads are associated with greater (average and maximum) order book variation, with the strongest correlation observed between level 5 relative spreads and variation measures.

Table 7. Correlation between spreads and order book variation. This table reports the correlation between the mean variation (first column) as well as the maximum variation (second column) and the level 1 relative spread (first row), the level 5 relative spread (second row), the level 10 relative spread (third row), the level 15 relative spread (forth row), as well as the level 20 relative spread (fifth row). All correlations are significantly different from zero at the 1%-significance level.

	Mean Variation	Max Variation
L1 Rel. Spread	−0.17 ***	−0.20 ***
L5 Rel. Spread	−0.22 ***	−0.26 ***
L10 Rel. Spread	−0.11 ***	−0.14 ***
L15 Rel. Spread	−0.15 ***	−0.18 ***
L20 Rel. Spread	−0.17 ***	−0.20 ***

*** indicates 0.01 significance level for a two-tailed test.

This result is intuitive. Larger spreads reflect higher trading costs, which deter investors from placing market orders and paying the associated liquidity premium. Instead, investors are more likely to submit limit orders in the face of limited liquidity. Conversely, tighter spreads increase the likelihood of order book variation, as consuming liquidity becomes cheaper, encouraging traders to use market orders more frequently.

These findings are consistent with related literature on equity markets, such as Ranaldo (2004), and align with the equilibrium model of liquidity supply and demand in limit order books proposed by Handa and Schwartz (1996).

5.2. Depth

Tables 8 and 9 present the results for ask depth and bid depth, respectively, across order book levels 1 to 20. By construction, depth is measured in units of the base currency. For instance, the level 1 ask depth in Table 8 for BTC on Binance is 0.329 BTC. Base currencies that are only available on a single exchange are excluded from the analysis, as they cannot be compared across exchanges.

The remaining base currencies are categorized into four groups: Bitcoin, Ether, US dollar and USD-pegged stablecoins. On average, Bitcoin and Ether exhibit greater depth on the bid side, while the US dollar and USD-pegged stablecoins show greater depth on the ask side.

For Bitcoin as the base currency, Binance and Kraken demonstrate higher depth across all order book levels compared to Huobi and OKEx. Although Binance and Kraken have similar depth values for the first and fifth price levels, Kraken exhibits greater depth at deeper levels of the order book. Kraken also shows high depth for Ether, whereas Binance has the lowest depth among the exchanges for Ether as the base currency. This indicates that Binance consolidates more liquidity in terms of depth for Bitcoin than for Ether. Interestingly, the large number of BTC trading pairs on Binance does not appear to negatively impact its depth.

Table 8. Ask side depth. This table reports the average level 1 ask depth (column 3), the average level 5 ask depth (column 4), the average level 10 ask depth (column 5), the average level 15 ask depth (column 6) and the average level 20 ask depth (column 7) in terms of the corresponding base currency (column 2) by the four exchanges (column 1) and the base currency. The numbers in the brackets report the standard errors of the corresponding averages.

Exchange	Base Currency	L1 Ask Depth	L5 Ask Depth	L10 Ask Depth	L15 Ask Depth	L20 Ask Depth
BINANCE	BTC	0.329 (0.0103)	2.1 (0.0548)	4.46 (0.111)	6.77 (0.166)	8.97 (0.212)
HUOBI	BTC	0.145 (0.0025)	1.19 (0.0322)	2.33 (0.0591)	3.37 (0.0762)	4.41 (0.0898)
KRAKEN	BTC	0.306 (0.0064)	2.08 (0.0405)	5.39 (0.122)	9.39 (0.218)	13.1 (0.341)
OKEX	BTC	0.0631 (0.002)	0.459 (0.0196)	0.917 (0.0439)	1.42 (0.0681)	1.99 (0.092)
BINANCE	ETH	0.879 (0.0076)	5.76 (0.0547)	12 (0.127)	17.7 (0.207)	23.8 (0.302)
HUOBI	ETH	2.35 (0.0537)	12 (0.329)	20.8 (0.485)	30.5 (0.757)	39.3 (0.969)
KRAKEN	ETH	3.55 (0.0816)	20.3 (0.706)	57.5 (2.02)	96.8 (3.53)	128 (4.65)
OKEX	ETH	0.91 (0.0244)	7.23 (0.379)	15.5 (0.858)	24.1 (1.35)	32.8 (1.88)
HUOBI	USD	4706 (522)	156,991 (28,672)	419,948 (71,680)	539,540 (78,370)	668,694 (84,430)
KRAKEN	USD	2307 (54.4)	15,743 (370)	45,004 (1142)	82,555 (2344)	118,909 (3596)
BINANCE	PAX	1114 (23.2)	9014 (300)	21,551 (804)	35,861 (1486)	50,158 (2155)
BINANCE	TUSD	2129 (69.2)	15,411 (583)	38,206 (1365)	54,017 (1925)	63,632 (2139)
BINANCE	USDC	1156 (40.7)	8338 (204)	18,984 (417)	31,411 (720)	43,418 (1101)
OKEX	USDK	686 (84.8)	5157 (754)	10,707 (1331)	16,443 (1681)	22,657 (2217)
BINANCE	USDT	1041 (16.3)	6818 (89.4)	14,958 (177)	23,130 (262)	31,490 (347)
HUOBI	USDT	963 (24.6)	6391 (175)	14,866 (406)	24,192 (623)	33,922 (802)
OKEX	USDT	332 (9.23)	2629 (95.6)	5645 (225)	8594 (339)	11,477 (430)

The comparison for USD as the base currency reveals patterns similar to those observed for spread measures. Huobi provides greater liquidity in terms of depth than Kraken on both the bid and ask sides. While Kraken shows similar depth on both sides of the book,

Huobi's ask side exhibits greater depth for the first and fifth price levels. The divergence in accumulated depth becomes more pronounced at deeper levels of the order book.

Table 9. Bid side depth. This table reports the average level 1 bid depth (column 3), the average level 5 bid depth (column 4), the average level 10 bid depth (column 5), the average level 15 bid depth (column 6) and the average level 20 bid depth (column 7) in terms of the corresponding base currency (column 2) by the four exchanges (column 1) and the base currency. The numbers in the brackets report the standard errors of the corresponding averages.

Exchange	Base Currency	L1 Bid Depth	L5 Bid Depth	L10 Bid Depth	L15 Bid Depth	L20 Bid Depth
BINANCE	BTC	0.392 (0.0151)	2.67 (0.0684)	5.15 (0.103)	7.02 (0.124)	8.47 (0.135)
HUOBI	BTC	0.146 (0.0025)	1.38 (0.0352)	2.74 (0.0622)	3.96 (0.079)	5.11 (0.0916)
KRAKEN	BTC	0.38 (0.0089)	2.49 (0.0532)	5.88 (0.102)	10.1 (0.176)	14.4 (0.29)
OKEX	BTC	0.0869 (0.0027)	0.733 (0.0274)	1.31 (0.044)	1.92 (0.0631)	2.6 (0.0878)
BINANCE	ETH	1.22 (0.007)	6.75 (0.054)	16.5 (0.129)	26.1 (0.206)	35 (0.289)
HUOBI	ETH	2.59 (0.0546)	18.4 (0.541)	29.9 (0.664)	40.9 (0.809)	50.7 (0.998)
KRAKEN	ETH	4.65 (0.118)	24.7 (0.652)	63.5 (1.95)	103 (3.28)	133 (4.22)
OKEX	ETH	1.15 (0.0335)	11.5 (0.485)	57.5 (20.2)	34.9 (1.55)	44.9 (2.1)
HUOBI	USD	3751 (717)	97,143 (17,995)	442,571 (73,177)	608,334 (87,762)	743,027 (97,323)
KRAKEN	USD	2428 (61.6)	15,532 (350)	40,661 (852)	72,862 (1473)	108,608 (2326)
BINANCE	PAX	1013 (27.2)	8190 (485)	19,226 (927)	31,993 (1500)	46,400 (2083)
BINANCE	TUSD	2067 (74.5)	14,601 (514)	35,812 (1216)	48,294 (1662)	56,376 (1930)
BINANCE	USDC	1380 (50.5)	8789 (246)	18,369 (440)	29,597 (696)	42,319 (1134)
OKEX	USDK	680 (82.9)	4422 (461)	10,286 (901)	15,643 (1331)	21,753 (1541)
BINANCE	USDT	995 (13.4)	7049 (89.2)	16,041 (191)	25,397 (286)	34,975 (375)
HUOBI	USDT	897 (21.8)	6145 (165)	14,595 (389)	23,742 (597)	32,837 (766)
OKEX	USDT	397 (10.1)	3644 (169)	7654 (299)	11,266 (416)	14,535 (508)

Stablecoins generally exhibit lower depth compared to their fiat currency counterpart, the USD. This result is intuitive, as investors may prefer trading in actual USD over its stablecoin alternatives. However, fiat trading pairs are relatively limited, and investors focusing exclusively on fiat as the base currency exclude a significant portion of cryptocurrencies from their investment options. For stablecoins, there is a negative relationship between the number of trading pairs and depth.

A comparison of these results with the spread measures suggests a positive relationship between depth and spread. Lower spreads are typically associated with higher depth across all order book levels in our sample. Additionally, we observe a negative relationship

between order book variation and depth for each base currency, with the exception of Huobi's USD trading pairs. Higher depth is generally associated with lower order book variation, a result that aligns with intuitive expectations.

5.3. Order Book Variation

Table 10 summarizes the results for order book variation, grouped by exchange and base currency type. The third column reports the daily average of the mean variation calculated over the 288 intraday observations, while the fourth column presents the daily average of the maximum variation experienced by an order book within a day. The values in parentheses represent the standard errors of the corresponding measures.

Table 10. Order book variation and return volatility by base type. This table reports the average mean variation (column 3), the average maximum variation (column 4) and the annualized return volatility by the four exchanges (column 1) and the base currency types (column 2). The numbers in the brackets report the standard errors of the corresponding averages.

Exchange	Base Type	Mean Variation	Max Variation	Return Volatility
BINANCE	cryptocurrency	1.97 (0.011)	12.5 (0.079)	132% (2.14%)
BINANCE	stablecoin	3.24 (0.034)	15.5 (0.203)	101% (2.61%)
HUOBI	cryptocurrency	2.95 (0.055)	19.3 (0.358)	147% (1.5%)
HUOBI	fiat currency	2.71 (0.055)	15.7 (0.409)	75% (3.3%)
HUOBI	stablecoin	6.86 (0.148)	39.4 (0.724)	103% (1.6%)
KRAKEN	cryptocurrency	1.63 (0.016)	10.5 (0.136)	84% (1.0%)
KRAKEN	fiat currency	2.32 (0.023)	14.3 (0.180)	114% (6.6%)
OKEX	cryptocurrency	1.48 (0.023)	8.04 (0.185)	145% (2.7%)
OKEX	stablecoin	2.02 (0.065)	13.0 (0.451)	138% (14.9%)

The mean (maximum) variation ranges from 1.48 (8.04) for OKEx's cryptocurrencies to 6.86 (39.4) for Huobi's stablecoins. The significant differences between the average means and maximums of intraday variation indicate high liquidity risk: the liquidity available at any given moment can vary substantially depending on the timing of an order submission. Periodic sharp drops in liquidity during a trading day pose risks for investors. Investors submitting *new* orders without up-to-date information on the state of the order book face heightened uncertainty, while *existing* limit orders are exposed to pick-off risk, which can be exploited by informed and fast traders.

Focusing on exchanges that offer currency pairs with stablecoins as base currencies (Binance, Huobi and OKEx), both the mean and maximum variation are higher for stablecoins compared to fiat currencies or cryptocurrencies. Among these exchanges, Huobi exhibits the highest variation for trades involving stablecoins. Kraken's data reveal higher variation for currency pairs with fiat currencies as base currencies compared to those with cryptocurrencies as base currencies. This suggests that, for Binance, Kraken and OKEx, investors provide more liquidity to cryptocurrencies than to fiat currencies and stablecoins—a surprising finding given the higher price volatility associated with base cryptocurrencies.

For Huobi, which offers all three types of base currencies, fiat currencies demonstrate the lowest variation among the base types. In contrast, OKEx and Binance, which do not offer fiat currencies, show lower variation for stablecoins and cryptocurrencies than the corresponding base types on Huobi. This suggests that, from an exchange's perspective, providing fiat currencies as base currencies may detract liquidity from stablecoins and other cryptocurrencies.

We further observe high return volatilities across exchanges and base types which are typical for cryptos. The volatilities range from 75% for fiat currencies as the base currency on Huobi to 145% for cryptocurrencies on OKEx. Comparing the results of order book variation and return volatility, we find that these two measures do not yield the same results across exchanges and base currencies, so order book variations and return volatility measure different aspects of liquidity.

Figure 1 shows the average mean variation and the average maximum variation on an intraday level. We aggregated the results on an hourly level instead of a daily level to study whether there are any intraday patterns. We can observe for all exchanges that the mean and maximum variation peak at hour 17 of the Coordinated Universal Time (UTC). This is best visible for the Huobi order books which exhibits the highest order book variation. Even though we find statistically significant differences between hour 17 and hour 16 as well as between hour 17 and hour 18, economical significance is rather small since the differences are between 0.3 and 1.4 order book ranks on average.

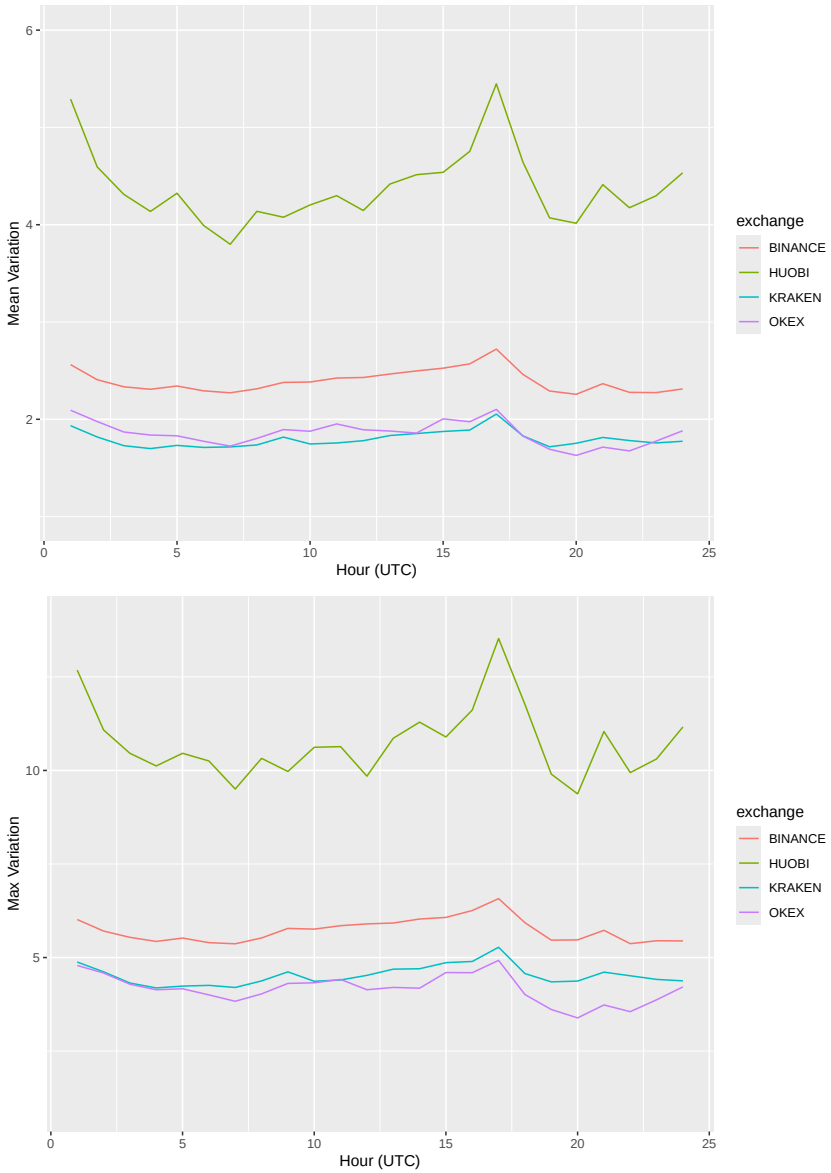


Figure 1. Intraday order book variation. These figures visualize the average mean variation (**top**) and average maximum variation (**bottom**) on the y-axis for each exchange indicated by the colors. The x-axis shows the hour during the day in Coordinated Universal Time (UTC).

5.4. Order Book Imbalance

Table 11 presents the normalized order book imbalances (NOBI), grouped by cryptocurrency exchanges and base currencies. The results show that the average NOBI is negative across all levels and base currencies for Binance, Kraken and OKEx. With the exception of Huobi's fiat currency, all exchanges and base currency types exhibit a preponderance of buying orders starting from level 10. This indicates that investors generally prefer to exchange their base currencies for target currencies.

Table 11. Normalized order book imbalance (NOBI) by exchange and base currency. This table reports the average normalized order book imbalance (NOBI) for level 1 (column 3), level 5 (column 4), level 10 (column 5), level 15 (column 6) and level 20 (column 7) by the exchanges (column 1) and base currency types (column 2). The numbers in the brackets are the standard errors of the corresponding average NOBIs.

Exchange	Base Type	L1 NOBI	L5 NOBI	L10 NOBI	L15 NOBI	L20 NOBI
BINANCE	cryptocurrency	−0.0742 (0.0010)	−0.1460 (0.0014)	−0.2120 (0.0016)	−0.2300 (0.0017)	−0.2250 (0.0017)
BINANCE	stablecoin	−0.0197 (0.0022)	−0.0008 (0.0022)	−0.0055 (0.0026)	−0.0070 (0.0027)	−0.0135 (0.0028)
HUOBI	cryptocurrency	0.0022 (0.0022)	−0.0955 (0.0032)	−0.1490 (0.0034)	−0.1660 (0.0035)	−0.1580 (0.0035)
HUOBI	fiat currency	0.0624 (0.0063)	0.0353 (0.0078)	0.0162 (0.0099)	0.0058 (0.0096)	0.0081 (0.0094)
HUOBI	stablecoin	0.0368 (0.0021)	0.0153 (0.0020)	−0.0080 (0.0026)	−0.0247 (0.0031)	−0.0276 (0.0035)
KRAKEN	cryptocurrency	−0.0578 (0.0033)	−0.0771 (0.0037)	−0.0691 (0.0044)	−0.0685 (0.0047)	−0.0760 (0.0049)
KRAKEN	fiat currency	−0.0267 (0.0025)	−0.0264 (0.0034)	−0.0086 (0.0035)	−0.0046 (0.0034)	−0.0163 (0.0034)
OKEX	cryptocurrency	−0.0128 (0.0037)	−0.2080 (0.0044)	−0.2760 (0.0048)	−0.2790 (0.0053)	−0.2580 (0.0058)
OKEX	stablecoin	−0.0353 (0.0051)	−0.1140 (0.0055)	−0.1490 (0.0058)	−0.1510 (0.0061)	−0.1400 (0.0063)

Huobi's fiat currency (USD) is the only base currency with a positive average order book imbalance across all levels, reflecting a preponderance of selling orders. However, beyond level 10, the imbalance is not significantly different from zero, suggesting that the order book becomes balanced at deeper levels. This aligns with our assumption that fiat currency pairs, such as USD, serve as entry and exit points to the cryptocurrency market.

In general, the best levels of the order book do not accurately represent the state of the entire order book. For Huobi, the level 1 NOBI is not significantly different from zero, indicating a balanced order book at the best level. However, for cryptocurrencies and stablecoins on Huobi, the NOBI turns negative starting at levels 5 and 10, respectively. Significant shifts in NOBI are also observed for other exchanges and base types. For instance, Binance's cryptocurrencies and both base currency types on OKEx show a sharp decline in NOBI at deeper levels. Additionally, Binance's stablecoins exhibit an insignificant level 5 NOBI.

These findings support previous literature and our results, highlighting that deeper levels of the order book provide more robust and reliable information compared to the best levels.

To compare order book imbalance with our variation measure, Table 12 reports the absolute normalized order book imbalance (ANOBI), grouped by cryptocurrency exchanges and base currencies. The results show that level 1 ANOBI values vary only slightly across base currency types and exchanges, ranging from 0.08 to 0.19. However, differences become more pronounced at deeper levels of the order book. ANOBI values increase significantly up to levels 10 and 15, stabilizing thereafter with no substantial changes by level 20, except for Huobi's stablecoins. These findings align with those in Table 12, confirming that deeper levels of the order book provide more reliable information about overall order book imbalance. This also supports the literature's assertion that the best levels are noisy and less informative.

Comparing the variation results in Table 10 with the level 15 ANOBI results in Table 12, we observe an overall negative relationship between the two variables. Higher order book imbalances are associated with lower variation risk. Notably, OKEx's stablecoins and other cryptocurrencies, as well as Binance's other cryptocurrencies, exhibit the three highest average ANOBIs and correspondingly account for three of the four lowest variation results. Conversely, Huobi's stablecoins and fiat currency, along with Binance's stablecoins, show the lowest average ANOBIs and rank among the four highest variation results.

Table 12. Absolute normalized order book imbalance (ANOBI) by exchange and base currency. This table reports the average absolute normalized order book imbalance (ANOBI) for level 1 (column 3), level 5 (column 4), level 10 (column 5), level 15 (column 6) and level 20 (column 7) by the exchanges (column 1) and base currency types (column 2). The numbers in the brackets are the standard errors of the corresponding average ANOBIs.

Exchange	Base Type	L1 ANOBI	L5 ANOBI	L10 ANOBI	L15 ANOBI	L20 ANOBI
BINANCE	cryptocurrency	0.119 (0.0008)	0.187 (0.0012)	0.249 (0.0014)	0.271 (0.0014)	0.275 (0.0013)
BINANCE	stablecoin	0.12 (0.0016)	0.121 (0.0016)	0.143 (0.0019)	0.155 (0.0019)	0.16 (0.0019)
HUOBI	cryptocurrency	0.0988 (0.0015)	0.148 (0.0027)	0.185 (0.0030)	0.202 (0.0030)	0.203 (0.0029)
HUOBI	fiat currency	0.104 (0.0047)	0.112 (0.0056)	0.142 (0.0067)	0.141 (0.0064)	0.136 (0.0065)
HUOBI	stablecoin	0.082 (0.0015)	0.0721 (0.0014)	0.0882 (0.0019)	0.104 (0.0023)	0.118 (0.0025)
KRAKEN	cryptocurrency	0.132 (0.0022)	0.151 (0.0025)	0.171 (0.0030)	0.18 (0.0033)	0.186 (0.0034)
KRAKEN	fiat currency	0.128 (0.0017)	0.168 (0.0023)	0.175 (0.0024)	0.166 (0.0023)	0.16 (0.0023)
OKEX	cryptocurrency	0.187 (0.0025)	0.286 (0.0032)	0.346 (0.0036)	0.358 (0.0038)	0.354 (0.0040)
OKEX	stablecoin	0.171 (0.0037)	0.216 (0.0038)	0.241 (0.0041)	0.255 (0.0042)	0.254 (0.0043)

The only exceptions are Huobi's and Kraken's other cryptocurrencies. Since higher order book imbalance is typically associated with higher transaction costs and lower liquidity, these results are both intuitive and consistent with theoretical expectations.

Next, we evaluate whether the bid and ask sides of the order book contribute differently to order book variation. Table 13 reports the correlations between the imbalance measures and the mean variation in the second column, as well as the maximum variation in the third column.

Panel (a) presents the results for the absolute normalized order book imbalance (ANOBI). The correlations are significantly negative across all order book levels. A noticeable jump in correlation occurs between the best level and the fifth level, with a decrease of –0.08 for both mean and maximum variation. Another smaller jump is observed from the fifth to the tenth level, with a decrease of –0.04 for mean variation only. These jumps suggest that the best levels are noisy and less informative. Beyond the tenth level, the results stabilize at approximately –0.3, indicating no substantial differences in the relationship between ANOBI and the two variation measures. This aligns with expectations: higher imbalance is associated with lower liquidity and higher trading costs, leading to fewer market order submissions. These findings are consistent with the overall results.

Panel (b) in Table 13 shows the correlations between the normalized order book imbalance (NOBI) and the variation measures. The correlations are significantly positive, ranging from 0.19 to 0.24 for levels 5 through 20. Similar to ANOBI, a large jump is observed from the first to the fifth level.

Table 13. Correlation of order book variation and imbalance. This table reports the correlation between the mean variation as well as the maximum variation average and the absolute normalized order book imbalance (ANOBI) in Panel (**a**) as well as the normalized order book imbalance (NOBI) in Panel (**b**) for the order book levels 1, 5, 10, 15 and 20 each. All correlations are significantly different from zero at the 1%-significance level.

(**a**) Absolute Normalized Order Book Imbalance			(**b**) Normalized Order Book Imbalance		
	Mean Variation	**Max Variation**		**Mean Variation**	**Max Variation**
L1 ANOBI	−0.19 ***	−0.21 ***	L1 NOBI	0.15 ***	0.13 ***
L5 ANOBI	−0.27 ***	−0.29 ***	L5 NOBI	0.21 ***	0.22 ***
L10 ANOBI	−0.31 ***	−0.31 ***	L10 NOBI	0.24 ***	0.23 ***
L15 ANOBI	−0.32 ***	−0.31 ***	L15 NOBI	0.24 ***	0.22 ***
L20 ANOBI	−0.32 ***	−0.30 ***	L20 NOBI	0.22 ***	0.19 ***

*** indicates 0.01 significance level for a two-tailed test.

Considering that order book imbalance is calculated as the accumulated ask volume minus the accumulated bid volume, there is strong evidence that greater depth on the bid side contributes more strongly to reduced variation. This is an intriguing result. As long as the order book provides sufficient depth on the bid side, enabling investors to sell their target currencies, the risk of variation tends to decrease. Conversely, lower bid-side depth is associated with higher variation risk. This relationship may help explain significant price drawdowns, as investors are more likely to sell their target currencies when they observe a decline in the volume of buy orders.

5.5. Regression Results for Order Book Variation

The results of the spread measures and order book imbalance suggest that variation is endogenous. We have outlined the negative relationship between variation and both spreads and imbalances, which is intuitive at first glance.

However, it is also plausible that higher variation could be associated with higher spreads and greater order book imbalance. Significant variation in the order book could lead to wider spreads, as larger gaps must be filled by new orders submitted to the order book. This process may take considerable time, causing the order book to maintain a large spread. Additionally, if order book variation were exogenous—that is, caused by factors outside the order book—it would occur independently of the current spread in the market. Nevertheless, the strong negative relationship we observe suggests a causal relationship from spreads to variation.

Similarly, variation could also drive order book imbalance. A large trade, for instance, may consume liquidity on one side of the order book, leading to imbalance. However, if variation were exogenous, it would occur independently of order book imbalance. The significantly negative relationship we observe implies a causal relationship from imbalance to variation.

We study the effects of spread, imbalance and their interaction on variation using panel regression models. We have an unbalanced panel since new listing and de-listings happened during the observed time horizon and some observations had to be ommitted due to inconsistencies as described in Section 3.2. Besides the pooled OLS model, we control for variables that are constant among the time but vary across trading pairs by including trading pair dummies ($D1_i, ..., DN_i$) and for variables that are constant across entities but vary over time can be done by including time dummies ($B1_t, ..., BT_t$). These models allows to eliminate bias from unobservables that change over time but are constant over trading pairs and vice versa. Trading pair fixed effects include, among others, the exchange they are traded on and the type of base currency. Time dummies account for, inter alia, the total market capitalization of the crypto market as well as crypto fear, greed and uncertainty.

$$Variation_{i,t} = \beta_1 Spread_{i,t} + \beta_2 ANOBI_{i,t} + \gamma_1 D1_i + ... + \gamma_n DN_i + \delta_1 B1_t + ... + \delta_T BT_t \quad (15)$$

This regression equation can be simplified to the following equation:

$$Variation_{i,t} = \beta_1 Spread_{i,t} + \beta_2 ANOBI_{i,t} + PairFixedEffects + TimeDummies \quad (16)$$

Based on the high correlations with variation reported in Tables 7 and 13, we select the level 5 VWAP relative spread and the level 10 ANOBI as key predictors. However, similar results are observed for other combinations of spread and imbalance levels. Some trading pairs did not have 10 limit orders outstanding during a day, 3386 observations (trading days) were excluded from the regression.

Table 14 presents the regression results for four types of models: pooled OLS, a trading pair fixed effects model, an OLS model with time dummies and a trading pair fixed effects model with time dummies. The coefficient signs align with expectations across all models, though the magnitudes vary. Higher spreads and order book imbalances are associated with lower variation risk. An interaction term is included to account for the collinearity between the level 5 VWAP relative spread and the level 10 ANOBI. While the interaction term is not significant in the pooled OLS model, it is significantly negative in the fixed effects models.

To determine the appropriate model, we perform an F-test for trading pair fixed effects and/or time dummies. The test results confirm the presence of significant individual and time effects, leading us to select the fixed effects model with both currency pair individual and time effects.

The regression results indicate that a one-percent increase in the level 5 VWAP relative spread decreases variation by 0.53% to 0.41%, depending on the level of the level 10 ANOBI. Similarly, a one-percent increase in the level 10 ANOBI leads to a variation change ranging from a decrease of 0.08% to an increase of 0.02%, depending on the level of the level 5 VWAP relative spread. Including the interaction term is essential to account for the correlation between spread and imbalance, as it significantly affects the results.

Economically, the effect of spread appears to be stronger than that of imbalance. The results provide robust evidence that variation can be explained by the selected liquidity measures. We hypothesize that large trades causing variation are strategically timed to coincide with periods of high liquidity in the currency pairs.

Table 14. Regression results: OLS vs fixed effects. This table reports regression results, the estimated coefficients with their corresponding standard errors in brackets and significance levels by stars, for four log-log models. Maximum variation (dependent variable) is regressed on the level 5 VWAP relative spread (row 1), the level 10 absolute normalized order book imbalance (ANOBI) (row 2) and the interaction term of the two explanatory variables (row 3). The models are a pooled OLS model (column 2), a trading pair fixed effects model (column 2), a model with time dummies (column 3), a trading pair fixed effects model with time dummies (column 4) and a trading pair fixed effects model with exchange and time dummies (column 5). In the bottom row, the results for a F test for trading pair fixed effects and/or timed dummies are presented in the last two rows.

	Dependent Variable:				
	log(Max Variation)				
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
log(L5 VWAP Spread)	−0.566 ***	−0.452 ***	−0.592 ***	−0.535 ***	−0.573 ***
	(0.005)	(0.007)	(0.004)	(0.007)	(0.008)
log(L10 ANOBI)	−0.035 ***	−0.069 ***	−0.050 ***	−0.061 ***	−0.067 ***
	(0.009)	(0.010)	(0.008)	(0.009)	(0.010)
log(L5 VWAP Spread): log(L10 ANOBI)	0.002	−0.009 ***	−0.003 **	−0.009 ***	−0.009 ***
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)
Binance					2.311 ***
					(0.103)
Huobi					2.547 ***
					(0.104)
Kraken					1.926 ***
					(0.132)
OKEx					1.736 ***
					(0.152)
Constant	−0.422 ***				
	(0.020)				
Trading Pair Fixed Effects (1081)	NO	YES ***	NO	YES ***	YES ***
Time Dummies (279)	NO	NO	YES ***	YES ***	YES ***
Observations	251,342	251,342	251,342	251,342	251,342
R^2	0.14	0.02	0.02	0.02	0.02
Adj. R^2	0.14	0.02	0.01	0.02	0.02
F Statistic	4474 ***	35.36 ***	32.87 ***	38.52 ***	42.87 ***

** and *** indicate 0.05, and 0.01 significance levels, respectively, for a two-tailed test.

Next, we investigate whether the results differ between negatively and positively imbalanced order books, as indicated in Section 5.4. We run two fixed effects regressions, including both individual and time effects, separately for negatively and positively imbalanced order books. The results are presented in Table 15, with column (1) reporting the results for negative imbalances and column (2) for positive imbalances.

The findings reveal that the effects of spreads and imbalances are stronger for negatively imbalanced order books than for positively imbalanced ones. These results support our hypothesis that higher depth on the bid side has a stronger contribution to reducing variation compared to higher depth on the ask side.

For order books with a preponderance of selling orders, the coefficient of the spread is smaller than that for negative imbalance. In contrast, the effect of imbalance itself does not differ significantly between negative and positive imbalances. This suggests that the observed differences in order book variation between negative and positive imbalances are primarily driven by spreads. The results further indicate that traders are willing to pay higher spreads to sell their currencies than to buy new ones.

Table 15. Regression results: negative imbalance vs positive imbalance. This table reports the results of two individual and time fixed effect models. Model (1) reports the regression results for observations with negative order book imbalances and model (2) reports the regression results for observations with positive order book imbalances. The estimated coefficients are presented with their corresponding standard errors in brackets and significance levels by stars.

	Dependent Variable:	
	log(Max Variation)	
	(1)	**(2)**
log(L5 VWAP Spread)	−0.537 ***	−0.487 ***
	(0.010)	(0.013)
log(L10 ANOBI)	−0.060 ***	−0.048 ***
	(0.012)	(0.014)
log(L5 VWAP Spread): log(L10 ANOBI)	−0.009 ***	−0.006 **
	(0.003)	(0.003)

** and *** indicate 0.05 and 0.01 significance levels, respectively, for a two-tailed test.

6. Conclusions

This paper has examined liquidity in cryptocurrency markets, focusing on order book data across four major exchanges: Binance, Kraken, Huobi and OKEx. Through an analysis of multiple liquidity measures, including spreads, order book depth, imbalance and variation, we offer new insights into the functioning and efficiency of these markets.

Our findings underscore the importance of spread and depth as liquidity measures. We observe that lower spreads are associated with higher depth across order book levels, reflecting greater liquidity. However, these relationships vary by base currency type and exchange. For example, fiat currencies on Huobi demonstrate remarkably flat spreads across order book levels compared to stablecoins or other cryptocurrencies. Similarly, depth measures reveal that Binance consolidates liquidity more effectively for Bitcoin trading pairs, while Kraken provides deeper order books for Ether and fiat-based trading pairs. These results highlight the distinct market structures and strategies employed by different exchanges.

We also introduce a novel measure of order book variation, capturing the stability of order books over time. Our analysis reveals a strong negative relationship between order book variation and both spreads and imbalances. These findings suggest that liquid markets exhibit greater stability, as trades are executed with minimal disruption to the order book. The variation measure further indicates that higher depth on the bid side reduces variation risk, emphasizing the role of buy-side liquidity in maintaining market stability. Conversely, significant variation often coincides with higher spreads, particularly during periods of low liquidity or sharp market movements.

Order book imbalance, another critical liquidity metric, provides valuable insights into the dynamics of buy and sell pressures. Our results indicate that imbalances tend to be more pronounced for certain base currency types and exchanges, with bid-side depth playing a pivotal role in mitigating variation risk. Notably, fiat currency trading pairs on Huobi demonstrate a unique imbalance pattern, suggesting their role as entry and exit points in cryptocurrency markets.

Our regression analysis supports the endogeneity of order book variation, with spreads and imbalances as key explanatory variables. Higher spreads and imbalances are consistently associated with lower variation risk, and the interaction between these factors further refines the explanatory power of our models. These findings align with theoretical expectations and contribute to the broader understanding of liquidity dynamics in cryptocurrency markets.

From a practical perspective, our findings have implications for various market participants. Investors can use these insights to optimize trading strategies by selecting exchanges and trading pairs with lower spreads and greater depth. Exchanges, in turn, can enhance liquidity by refining market design, such as through the introduction of designated liquidity providers or the adjustment of trading pair offerings. Finally, regulators can leverage these results to evaluate market efficiency and assess the impact of fragmentation within the crypto ecosystem.

While this analysis provides robust insights into cryptocurrency market liquidity, it also has limitations. The study relies on data from a limited timeframe and focuses on four major exchanges, which may not capture the full diversity of the cryptocurrency market. Additionally, the measures and models used, though comprehensive, do not account for external factors such as macroeconomic events or regulatory changes that may influence liquidity. The use of order book data, while detailed, may also introduce biases due to differences in reporting standards across exchanges. Borri and Shakhnov (2022) further outline the issue of fake volume and/or noneconomic wash trading on crypto exchanges. Even though we solely used order book data and did not rely on any trading data, we nevertheless do not know if the practices at the times also had an impact on our results. Future studies could address these limitations by incorporating longer time horizons, more exchanges and broader market conditions.

In conclusion, this study demonstrates that liquidity measures derived from order book data provide a comprehensive understanding of market dynamics in the cryptocurrency space. The unique 24/7 trading nature of these markets, combined with their diverse trading pair structures, presents both challenges and opportunities for liquidity analysis. Future research could extend these findings by exploring alternative liquidity measures or investigating the impact of external events on order book dynamics. Our results lay the groundwork for such inquiries, contributing to the ongoing development of efficient and transparent cryptocurrency markets.

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